

Learning Camera Geometry from Unlabeled Real-World Dynamic Video

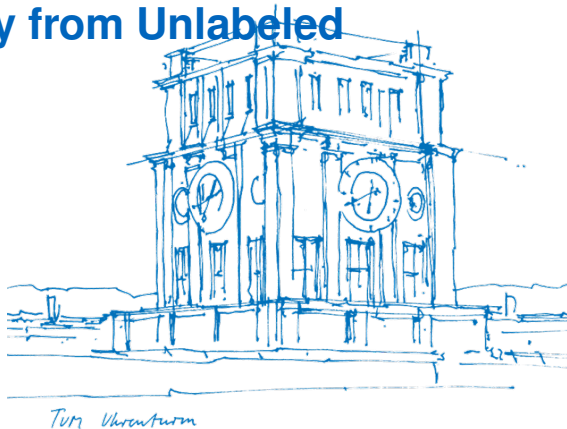
Master's Thesis Defense

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Outline

1. Motivation & Background
2. Existing Models: AnyCam & AnyCalib
3. Our Framework: Fusing Calibration & Pose
4. Contributions
5. Training Setup & Losses
6. Results
7. Conclusion & Future Work

Motivation

Recover **camera poses** (R, t) and **calibration** (K) from **unlabeled casual video**

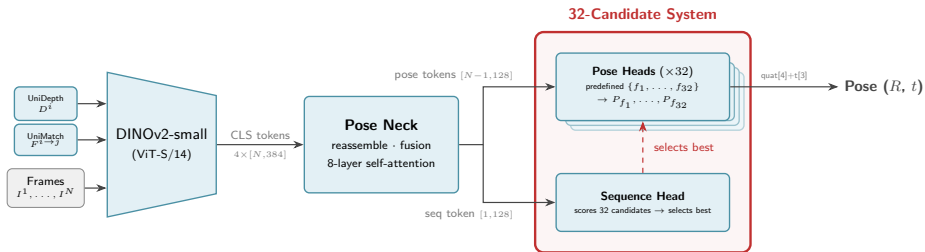
Why is this hard?

- Casual video: dynamic objects, unknown cameras, no 3D labels
- Classical SfM/SLAM fails on dynamic scenes
- Supervised methods need expensive ground truth that doesn't scale

What exists today:

- Strong pretrained models for individual sub-tasks: depth, optical flow, calibration, pose
- But they operate **in isolation** — no joint reasoning, no temporal consistency

The Calibration Problem in AnyCam



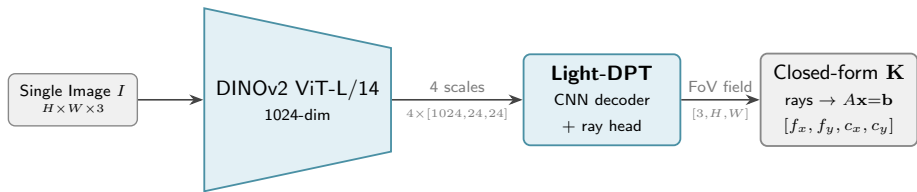
32-candidate focal length system:

- Generates 32 focal length guesses
- Runs flow reprojection for each
- Sequence head selects the best

AnyCam calibration error:

Dataset	Focal length MAPE
Sintel	45.1%
TUM-RGBD	63.7%

AnyCalib — A Path to Better Calibration

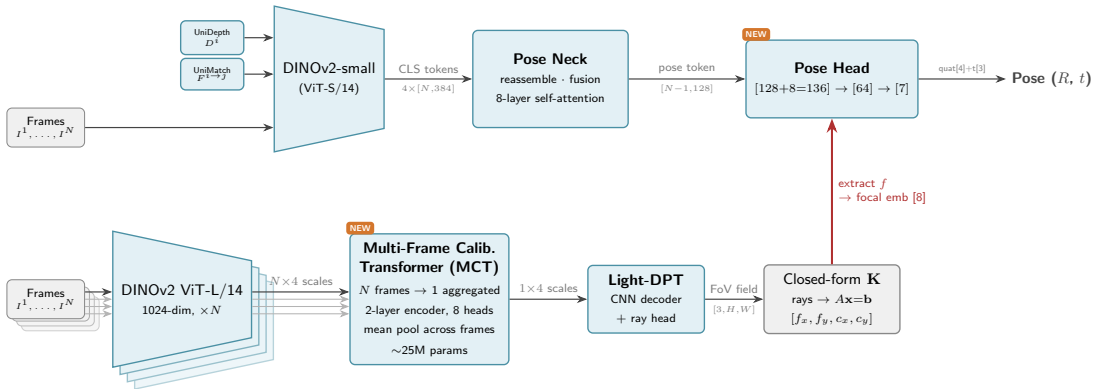


AnyCalib calibration error:

Dataset	AnyCam	AnyCalib
Sintel	45.1%	30.7%
TUM-RGBD	63.7%	11.7%

- Per-frame only — noisy, inconsistent
- Can we enforce **multi-frame consistency**?
- Can we improve on AnyCalib too?

Our Framework — Fusing Calibration & Pose



Contributions

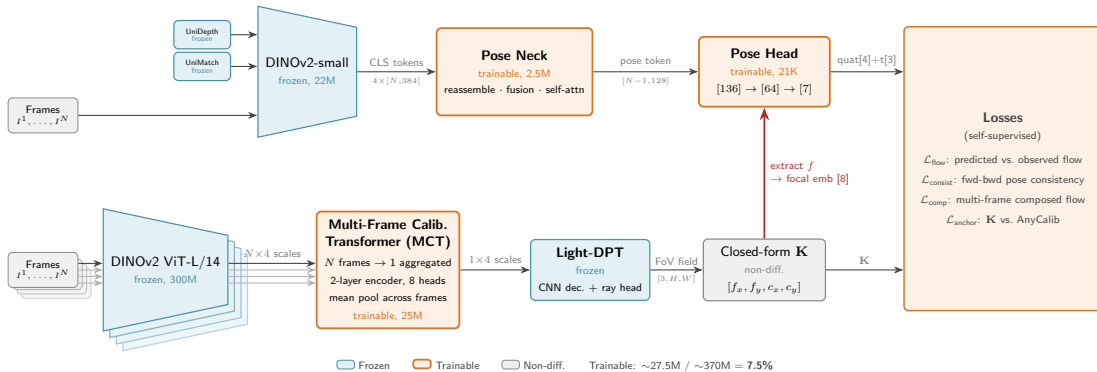
1. Multi-Frame Calibration Transformer (MCT)

- Cross-frame attention on calibration backbone features (4 scales)
- Aggregates N frames $\rightarrow$ 1 consistent calibration
- Architecture-agnostic: inserts between any backbone and decoder
- Only ~ 25 M trainable params (7.5% of total)

2. Demonstrated on AnyCam + AnyCalib

- Fuses a self-supervised pose estimator with a calibration network
- Improves *both* pose accuracy and calibration accuracy
- Fully self-supervised — no ground truth labels

Training Setup



Training Losses

Flow Reprojection (self-supervised, main signal):

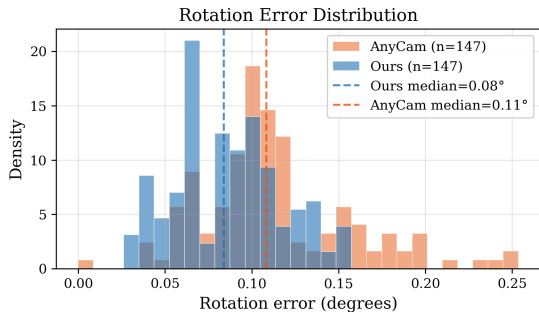
$$\mathbf{x}' = \pi \left(\mathbf{K} \left(R \mathbf{K}^{-1} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} d + \mathbf{t} \right) \right), \quad \mathcal{L}_{\text{flow}} = \|\mathbf{x}' - \mathbf{x} - \mathbf{w}_{\text{obs}}\|$$

Calibration Anchor — why $\mathbf{K}$ needs constraints:

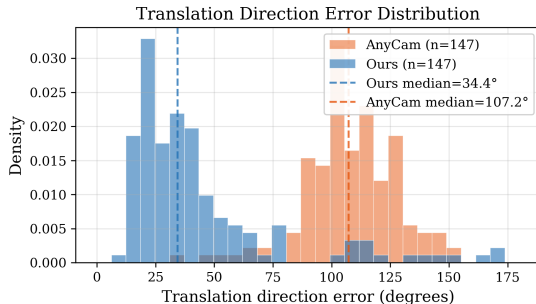
$$R \approx I, \mathbf{t} \approx 0 \implies \mathbf{K}^{-1}(d \mathbf{K} \mathbf{x} + \mathbf{t}) \approx \mathbf{x} \quad (\text{any } \mathbf{K} \text{ works})$$

$$\mathcal{L}_{\text{anchor}} = \|\mathbf{K}_{\text{pred}} - \mathbf{K}_{\text{AnyCalib}}\|$$

Results — Pose Error

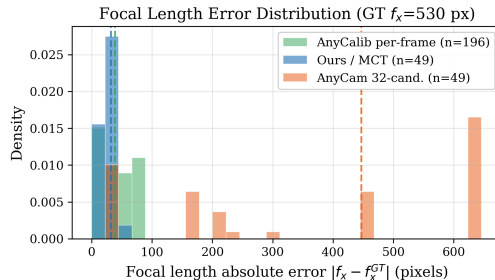
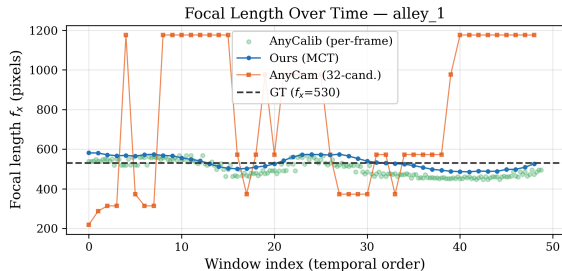


Rotation error	Mean	Median
AnyCam	0.121°	0.108°
Ours	0.086°	0.084°



Translation error	Mean	Median
AnyCam	108.5°	107.2°
Ours	44.3°	34.4°

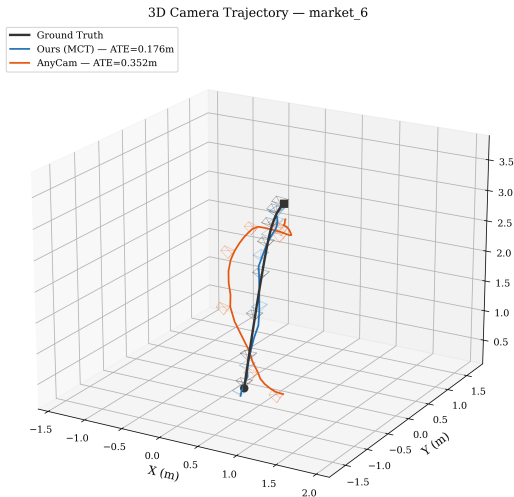
Results — Calibration Error



Calibration error	MAE (px)	MAPE
AnyCam	362.1	68.2%
AnyCalib	40.4	7.6%
Ours	28.5	5.4%

* *Sintel alley_1* sequence

Trajectory Visualization — Sintel market_6



Conclusion & Future Work

Summary:

- Proposed a framework for fusing pretrained calibration & pose models
- MCT: multi-frame calibration consistency via feature-level aggregation
- Demonstrated on AnyCam + AnyCalib: **28% better translation, 16% better calibration**

Future work:

- Longer sequences & non-pinhole camera models
- Joint depth refinement with calibration and pose

Thank you! Questions?